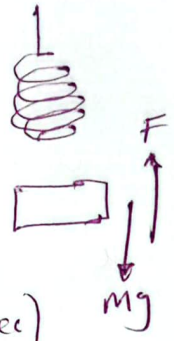
Maglev Modeling

① Physical model:

m = 0.1kg (mass of levitated object)
z = height (gap b/w magnet & object) [m]
i = current through electromagnetic (A)
u = applied voltage [V] : Control input
R = 2Ω (coil resistance) L = 0.05H (coil inductance)
Magnetic force $f = k \cdot i^2/z^2$ (k = force constant)



② Governing Equations:

Electric Circuit (coil) : Applied voltage $u = L \cdot (di/dt) + R \cdot i$
Mechanical motion : $m \cdot \ddot{z} = mg - F$

③ State Space:

$\dot{x} = f(x, u)$

$x = [x_1, x_2, x_3]$ x_1 = position, x_2 = velocity, x_3 = current

$u = [u, d]$ u = applied voltage, d = disturbance

Assumptions
$d = 0$
L = Constant
Friction = 0

To drive state space matrix

$\dot{x}_1 = x_2$ ——— ①

From Mechanical Equation

$m\ddot{x} = mg - \frac{ki^2}{x^2}$

$\ddot{x} = \dot{x}_2$ & $x = x_1$, $i = x_3$

$\dot{x}_2 = g - \frac{kx_3^2}{mx_1^2}$ ——— ②

From Electric equation

$$U = L\frac{di}{dt} + Ri$$

$$\frac{di}{dt} = \dot{x}_3 \quad \& \quad i = x_3$$

$$U = L\dot{x}_3 + Rx_3$$

$$\dot{x}_3 = -\frac{Rx_3}{L} + \frac{U}{L} \quad \text{---} (3)$$

Recall $\dot{x} = Ax + BU$

Therefore $\dot{x}_1 = x_2$ --- (1)

$$\dot{x}_2 = -\frac{kx_3^2}{Mx_1^2} + g \quad \text{---} (2)$$

$$\dot{x}_3 = -\frac{Rx_3}{L} + \frac{U}{L} \quad \text{---} (3)$$

We have ~~A = [0] $\dot{x}$~~

(4) Linearization:

At equilibrium $x_1 = z_0$, $x_2 = 0$, $x_3 = i_0$ & $U = U_0$

From equation (2)

$$0 = -\frac{ki_0^2}{Mz_0^2} + g \Rightarrow i_0^2 = \frac{Mgz_0^2}{k} \Rightarrow i_0 = z_0\sqrt{\frac{Mg}{k}} \quad (4)$$

From equation (3)

$$0 = -\frac{Ri_0}{L} + \frac{U_0}{L} \Rightarrow U_0 = Ri_0 \quad \text{---} (5)$$

from equation (2)

$$0 = -\frac{ki_0^2}{Mz_0^2} + g \Rightarrow Mg = \frac{ki_0^2}{z_0^2}$$

Substitute [$i_0 = 0.7$ and $z_0 = 0.05M$] and other parameters

$$U_0 = Ri_0 = 1.4V$$

$$K = \frac{Mgz_0^2}{i_0^2} = \frac{0.1 \times 9.8 \times (0.05)^2}{(0.7)^2} = 0.005\ kgm^3/A^2s^2$$

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To linearize the system $\dot{x} = f(x, u)$, we compute Jacobian Matrices at equilibrium point (x_0, u_0).

$$A = \left.\frac{\partial f}{\partial x}\right|_{x_0, u_0} = \begin{bmatrix} \frac{\partial \dot{x}_1}{\partial x_1} & \frac{\partial \dot{x}_1}{\partial x_2} & \frac{\partial \dot{x}_1}{\partial x_3} \\ \frac{\partial \dot{x}_2}{\partial x_1} & \frac{\partial \dot{x}_2}{\partial x_2} & \frac{\partial \dot{x}_2}{\partial x_3} \\ \frac{\partial \dot{x}_3}{\partial x_1} & \frac{\partial \dot{x}_3}{\partial x_2} & \frac{\partial \dot{x}_3}{\partial x_3} \end{bmatrix}_{x_0, u_0}$$

Using eqn 1, 2 and 3.

$$\frac{\partial \dot{x}_1}{\partial x_1} = 0 \quad \frac{\partial \dot{x}_1}{\partial x_2} = 1 \quad \frac{\partial \dot{x}_1}{\partial x_3} = 0$$

$$\frac{\partial \dot{x}_2}{\partial x_1} = \frac{2kx_3^2}{Mx_1^3} \quad \frac{\partial \dot{x}_2}{\partial x_2} = 0 \quad \frac{\partial \dot{x}_2}{\partial x_3} = -\frac{2kx_3}{Mx_1^3}$$

$$\frac{\partial \dot{x}_3}{\partial x_1} = 0 \quad \frac{\partial \dot{x}_3}{\partial x_2} = 0 \quad \frac{\partial \dot{x}_3}{\partial x_0} = -\frac{R}{L}$$

at equilibrium

$$A = \begin{bmatrix} 0 & 1 & 0 \\ \frac{2ki_0^2}{Mz_0^3} & 0 & -\frac{2ki_0}{Mz_0^2} \\ 0 & 0 & -R/L \end{bmatrix}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{x_0, u_0} = \begin{bmatrix} \frac{\partial \dot{x}_1}{\partial u} \\ \frac{\partial \dot{x}_2}{\partial u} \\ \frac{\partial \dot{x}_3}{\partial u} \end{bmatrix}_{x_0, u_0} \Rightarrow B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \end{bmatrix}$$

Substituting equilibrium parameters

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 3924 & 0 & -28.03 \\ 0 & 0 & -40 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ -40 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 20 \end{bmatrix}$$

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Final linearized State - space model

$$\dot{\Delta x} = \begin{bmatrix} 0 & 1 & 0 \\ 392.4 & 0 & -28.03 \\ 0 & 0 & -40 \end{bmatrix} \Delta x + \begin{bmatrix} 0 \\ 0 \\ 20 \end{bmatrix}$$

(5) To check stability:

Find open loop Eigen values:

$$\det(A - \lambda I) = 0 \qquad I = \text{Identity matrix}$$

Substituting A and I

$\lambda_1 = +19.81$ $\lambda_2 = -19.81$ $\lambda_3 = 40$

$\Rightarrow$ Unstable System.

(6) To check Controllability:

$$C = [B \; AB \; A^2B]$$

$$AB = \begin{bmatrix} 0 & 1 & 0 \\ 392.4 & 0 & -28.03 \\ 0 & 0 & -40 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 20 \end{bmatrix} = \begin{bmatrix} 0 \\ -560.6 \\ -800 \end{bmatrix}$$

$$A^2B = A(AB) = \begin{bmatrix} -560 \\ 22423 \\ 32000 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & -560.6 \\ 0 & -560.6 & 22423 \\ 20 & -800 & 32000 \end{bmatrix}$$

$$\text{Det}(C) = -6.28 \times 10^6$$

Since $|C| \neq 0$, System is fully Controllable

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PID Analytical Design:

Recall: $A = \begin{bmatrix} 0 & 1 & 0 \\ 392.4 & 0 & -28.03 \\ 0 & 0 & -40 \end{bmatrix}$ $B = \begin{bmatrix} 0 \\ 0 \\ 20 \end{bmatrix}$

$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$

(1) To find Transfer function (G(s))

$G(s) = C[SI - A]^{-1} \cdot B$ I = Identity Matrix

$$SI - A = \begin{bmatrix} S & 0 & 0 \\ 0 & S & 0 \\ 0 & 0 & S \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 392.4 & 0 & -28.03 \\ 0 & 0 & -40 \end{bmatrix} = \begin{bmatrix} S & -1 & 0 \\ -392.4 & 0 & 28.3 \\ 0 & 0 & S+40 \end{bmatrix}$$

$$\det[SI - A] = S[s(s+40)] + 1[-392.4(s+40)]$$
$$= s^3 + 40s^2 - 392.4s - 15696$$

To find Cofactor matrix:

$$C_{11} = \begin{vmatrix} S & 28.03 \\ 0 & S+40 \end{vmatrix} = s^2 + 40s$$

$$C_{12} = \begin{vmatrix} -392.4 & 28.03 \\ 0 & S+40 \end{vmatrix} = -392.4(S+40)$$

$$C_{13} = \begin{vmatrix} -392.4 & S \\ 0 & 0 \end{vmatrix} = 0$$

$$C_{21} = \begin{vmatrix} -1 & 0 \\ 0 & S+40 \end{vmatrix} = S+40$$

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$$C_{22} = \begin{vmatrix} s & 0 \\ 0 & s+40 \end{vmatrix} = s(s+40) = s^2+40s$$

$$C_{23} = \begin{vmatrix} s & -1 \\ 0 & 0 \end{vmatrix} = 0$$

$$C_{31} = \begin{vmatrix} -1 & 0 \\ s & 28.03 \end{vmatrix} = -28.03$$

$$C_{32} = \begin{vmatrix} s & 0 \\ 392.4 & 28.03 \end{vmatrix} = 28.03s$$

$$C_{33} = \begin{vmatrix} s & -1 \\ -392.4 & s \end{vmatrix} = s^2 - 392.4$$

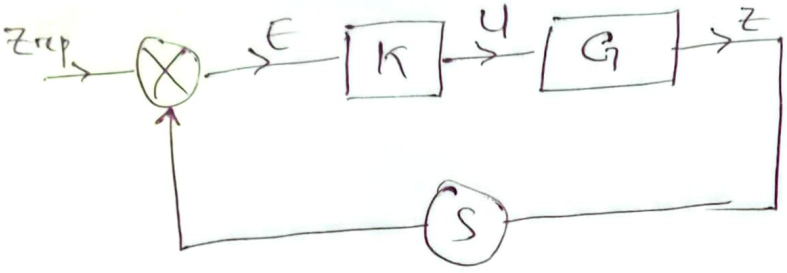
$$C^T = \begin{bmatrix} s^2+40s & s+40 & -28.03 \\ -392.4(s+40) & s^2+40s & 28.03s \\ 0 & 0 & s^2-392.4s \end{bmatrix}$$

$$\therefore G(s) = \frac{1}{s^3+40s^2-392.4s-15696} \cdot \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} s^2+40s & s+40 & -28.03 \\ -392.4(s+40) & s^2+40s & 28.03s \\ 0 & 0 & s^2-392.4s \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 20 \end{bmatrix}$$

$$G(s) = \frac{-560.6}{s^3+40s^2-392.4s-15696}$$

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PID Close loop



$E = Z_{ref} - Z$ ---(1)

$U(s) = k \cdot E(s)$ ---(2)

$Z = G \cdot U$ ---(3)

from above equations:

$Z = G \cdot kE(s) \Rightarrow G \cdot k(Z_{ref} - Z)$

$Z = GkZ_{ref} - GkZ \Rightarrow Z(1+Gk) = GkZ_{ref}$

$$T.f = G.s = \frac{\text{output}}{\text{input}} = \frac{Z}{Z_{ref}} = \frac{Gk}{1+Gk}$$

$1 + Gk = 0$ gives poles of the close loop.

Substuting $G(s)$ and $k = kp + kds + \frac{ki}{s}$

$$1 + \left[kp + kds + \frac{ki}{s}\right] \cdot \left[\frac{-560.6}{s^3 + 40s^2 - 392.4s - 15696}\right] = 0$$

$$\Rightarrow s(s^3 + 40s^2 - 392.6s - 15696) = -560.6(kps + kds^2 + ki)$$

$$\Rightarrow s^4 + 40s^3 - 392.4s^2 - 15696s = -560.6kps + (-560.6kds^2) - 560.6ki$$

$$\Rightarrow s^4 + 40s^3 + (-392.4 - 560.6kd)s^2 + (-15696 - 560.6kp)s - 560.6ki = 0$$

Comparing above with general form:

$a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} \cdots\cdots a_n = 0$

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we have

$$a_2 = -392.4 - 560.6k_d$$

$$a_3 = -15696 - 560.6k_p$$

for stability, Using Routh - Hurwitz Rules:

$a_2 > 0 \Rightarrow k_d > 0.70$

$a_3 > 0 \Rightarrow k_p > 27.998$

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